CRITICAL PROFILE DECOMPOSITIONS AND THE RADIATIVE AND MULTIBUBBLE ALTERNATIVES FOR NAVIER–STOKES BLOWUP

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Abstract. We prove a concentration-compactness reduction for critical data sequences arising from Navier–Stokes blowup rescalings, under explicitly stated analytic hypotheses. These hypotheses are formulated in the standard language of rescaled suitable weak solutions, critical profile decompositions in $L^3_\sigma(\mathbb{R}^3)$, Kato small-data theory, exterior regularity away from a fixed blowup point, and nonlinear profile stability on compact parabolic cylinders. Profiles below the small-data threshold are perturbative, profiles separated from the blowup point are regular near that point, loss of critical tightness is detected by a non-vanishing profile or remainder at infinity, and finite single-center or distinct-center multibubble configurations reduce to single-profile blowup sequences modulo errors vanishing in the local distributional equation. Consequently, no radiative or multibubble alternative remains unless one of the stated analytic hypotheses fails.

1. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

on $\mathbb{R}^3 \times (0, T)$, where $\nu > 0$. We fix a candidate singular point (x_*, T) and study sequences $t_n \uparrow T$, $x_n \to x_*$, $\lambda_n \downarrow 0$. The associated rescaled solutions are

$$v_n(y, s) = \lambda_n u(x_n + \lambda_n y, t_n + \lambda_n^2 s), \qquad q_n(y, s) = \lambda_n^2 p(x_n + \lambda_n y, t_n + \lambda_n^2 s). \tag{1}$$

The associated data at rescaled time $s = 0$ are denoted by

$$f_n(y) = v_n(y, 0) = \lambda_n u(x_n + \lambda_n y, t_n).$$

The critical norm is $\|f\|_{L^3(\mathbb{R}^3)}$. We assume throughout that

$$\sup_n \|f_n\|_{L^3(\mathbb{R}^3)} \le M < \infty. \tag{2}$$

The analytic hypotheses used below are stated explicitly. The critical profile decomposition, the nonlinear stability statement, and the single-profile Liouville theorem enter as hypotheses or cited analytic results. The argument derives their consequences for radiative tails and multibubble configurations. In particular, whenever a single-profile blowup sequence is excluded below, that step is attributed to the single-profile rigidity hypothesis.

The main result is Theorem 7.1. It states that if a bounded critical data sequence has the standard profile expansion, if profiles below the small-data threshold are perturbative, if profiles away from the blowup point are regular near that point, if nonlinear profile interactions vanish on compact parabolic cylinders, and if the remaining single-profile sequences are excluded by the relevant rigidity theorem, then no radiative or multibubble alternative remains. If such an alternative occurs, one of these analytic hypotheses has failed.

The proof follows the concentration-compactness reduction in five steps. First, the critical profile expansion and L^3-decoupling give only finitely many non-perturbative profiles above the Kato threshold. Second, Kato small-data theory and exterior regularity exclude, respectively, subthreshold profiles and profiles separated from the blowup point. Third, compactness together with a vanishing critical remainder implies uniform critical tightness. Fourth, nonlinear stability on compact parabolic cylinders shows that locally vanishing secondary profiles change the equation only by an error tending to zero in $L^1_s H^{-1}_y$. Finally, one-center scale separation and distinct-center configurations reduce to a single-profile blowup sequence, where the rigidity hypothesis applies.

2. Critical Profiles

We first record the critical profile assumptions and two elementary consequences that will be used repeatedly.

Lemma 2.1 (Critical scaling). *For $\lambda > 0$, $x_0 \in \mathbb{R}^3$, and $f \in L^3(\mathbb{R}^3)$,*

$$g(x) = \lambda^{-1} f\left(\frac{x - x_0}{\lambda}\right)$$

satisfies

$$\|g\|_{L^3(\mathbb{R}^3)} = \|f\|_{L^3(\mathbb{R}^3)}.$$

Proof. This is the change of variables $y = (x - x_0)/\lambda$:

$$\int_{\mathbb{R}^3} |g(x)|^3 dx = \int_{\mathbb{R}^3} \lambda^{-3} \left| f\left(\frac{x - x_0}{\lambda}\right)\right|^3 dx = \int_{\mathbb{R}^3} |f(y)|^3 dy.$$

□

Assumption 2.2 (Critical profile decomposition). After passing to a subsequence, the bounded sequence (f_n) has the following profile expansion. There are divergence-free profiles $\phi^j \in L^3(\mathbb{R}^3)$, scales $\lambda_{j,n} > 0$, centers $x_{j,n} \in \mathbb{R}^3$, and remainders r_n^J such that, for every finite J,

$$f_n(x) = \sum_{j=1}^{J} (\lambda_{j,n})^{-1} \phi^j \left(\frac{x - x_{j,n}}{\lambda_{j,n}}\right) + r_n^J(x). \tag{3}$$

The parameters are pairwise orthogonal:

$$\frac{\lambda_{i,n}}{\lambda_{j,n}} + \frac{\lambda_{j,n}}{\lambda_{i,n}} + \frac{|x_{i,n} - x_{j,n}|}{\max(\lambda_{i,n}, \lambda_{j,n})} \longrightarrow \infty \qquad (i \neq j), \tag{4}$$

and the L^3 mass decouples:

$$\|f_n\|_3^3 = \sum_{j=1}^{J} \|\phi^j\|_3^3 + \|r_n^J\|_3^3 + o_n(1). \tag{5}$$

The linear heat evolution of the remainder is small in the critical stability topology used for Navier–Stokes perturbation theory, and the corresponding nonlinear stability statement holds on compact parabolic time intervals. Finally, if a residual sequence has nonzero local L^3 mass after the listed profiles and the perturbative remainder have been accounted for, then a further subsequence produces an additional nonzero profile at the observed center and scale.

Remark 2.3. This is the standard concentration-compactness hypothesis used in critical Navier–Stokes profile arguments; see Kato's critical theory [Kat84] and Gallagher's Navier–Stokes profile decomposition [Gal01]. This is the usual completeness condition: no nonzero local critical mass remains outside the extracted profiles and the perturbative remainder.

Lemma 2.4 (Finite profile decoupling). *Let $\phi^1, \ldots, \phi^N \in L^3(\mathbb{R}^3)$, and suppose the associated scales and centers satisfy* (4)*. Then*

$$\left\| \sum_{j=1}^{N} (\lambda_{j,n})^{-1} \phi^j \left(\frac{x - x_{j,n}}{\lambda_{j,n}}\right)\right\|_3^3 = \sum_{j=1}^{N} \|\phi^j\|_3^3 + o_n(1).$$

Proof. For smooth compactly supported profiles, the proof is an induction on N. After rescaling by the N-th scale and center, each of the first $N-1$ profiles either has vanishing amplitude on compact sets, concentrates on a set of vanishing measure, or escapes to spatial infinity. Hence the sum of the first $N-1$ profiles converges pointwise almost everywhere to zero in the coordinates of the N-th profile and remains bounded in L^3. The Brezis–Lieb lemma [BL83] separates the N-th profile from that sum. The induction hypothesis treats the remaining profiles. The general L^3 case follows by approximating the profiles by smooth compactly supported functions and using Lemma 2.1. ☐

Lemma 2.5 (Only finitely many large profiles)**.** *Let $\eta > 0$. Under (2) and Assumption 2.2, the number of profiles satisfying $\|\phi^j\|_3 \geq \eta$ is at most*

$$\left\lfloor \frac{M^3}{\eta^3} \right\rfloor.$$

Proof. For every finite set J of such profiles, (5) gives

$$M^3 \geq \limsup_{n\to\infty} \|f_n\|_3^3 \geq \sum_{j\in J} \|\phi^j\|_3^3 \geq (\#J)\eta^3.$$

The asserted bound follows. ☐

3. Perturbative and Exterior Profiles

The two standard ways in which a profile is excluded from the analysis near the blowup point are small critical norm and positive physical separation from x_*.

Theorem 3.1 (Kato small-data theorem)**.** *There exists $\varepsilon_0 > 0$ such that, if $a \in L^3_\sigma(\mathbb{R}^3)$ and $\|a\|_3 \leq \varepsilon_0$, then the Navier–Stokes equations with initial data a have a unique global mild solution satisfying*

$$\sup_{t>0} \|u(t)\|_3 + \sup_{t>0} t^{1/8} \|u(t)\|_4 \leq C\|a\|_3. \tag{6}$$

Moreover the solution map is Lipschitz in this critical norm on sufficiently small balls, and the nonlinear part is quadratic:

$$\sup_{0<t<T} \|u(t) - e^{\nu t\Delta}a\|_3 + \sup_{0<t<T} t^{1/8}\|u(t) - e^{\nu t\Delta}a\|_4 \leq C_T\|a\|_3^2$$

for every finite T.

Proof. This is the classical fixed-point theorem of Kato [Kat84]. The proof uses the heat estimate

$$\|e^{\nu t\Delta}a\|_3 + t^{1/8}\|e^{\nu t\Delta}a\|_4 \leq C\|a\|_3$$

and the bilinear estimate for

$$-\int_0^t e^{\nu(t-s)\Delta}\mathbb{P}\nabla\cdot(u\otimes v)(s)ds.$$

Here $\mathbb{P}$ denotes the Leray projection. The latter estimate follows from the heat kernel bound for $e^{\nu(t-s)\Delta}\mathbb{P}\nabla\cdot$, Hölder's inequality, and the convergent integrals $\int_0^t (t-s)^{-3/4}s^{-1/4}ds$ and $\int_0^t (t-s)^{-7/8}s^{-1/4}ds$. A contraction on the ball of radius $2C\|a\|_3$ gives the solution and the stated bounds. ☐

Corollary 3.2 (Small profiles are perturbative)**.** *Every profile ϕ^j satisfying $\|\phi^j\|_3 \leq \varepsilon_0$ generates a global perturbative Navier–Stokes profile. Such a profile cannot be a non-perturbative blowup profile in the perturbative class considered here. Consequently every non-perturbative profile satisfies*

$$\|\phi^j\|_3 > \varepsilon_0.$$

Proof. Apply Theorem 3.1. The estimate (6) is invariant under Navier–Stokes scaling, so every rescaled copy of the small solution remains uniformly perturbative on compact rescaled time intervals. □

Assumption 3.3 (Exterior regularity)**.** The point (x_*, T) is the only possible singular point under consideration. For every $r > 0$, every $R < \infty$, and every integer $k \geq 0$, there are $\delta > 0$ and $C_{r,R,k} < \infty$ such that

$$\sup_{T-\delta<t<T} \Big(\|u(t)\|_{C^k(\overline{B_R} \setminus B_r(x_*))} + \|p(t)\|_{C^k(\overline{B_R} \setminus B_r(x_*))} \Big) \leq C_{r,R,k}.$$

Lemma 3.4 (Divergence-free localization near the blowup point)**.** *Assume Assumption 3.3. Let $r_0 > 0$, and let $\chi \in C_c^\infty(B_{2r_0}(x_*))$ satisfy $\chi \equiv 1$ on $B_{r_0}(x_*)$. There is a time-dependent vector field w, supported in $B_{2r_0}(x_*) \setminus B_{r_0}(x_*)$, such that*

$$\nabla \cdot w = \nabla \chi \cdot u.$$

Consequently

$$u_{\text{near}} = \chi u - w, \qquad u_{\text{far}} = u - u_{\text{near}}$$

are divergence-free. Moreover $u_{\text{near}} = u$ on $B_{r_0}(x_*)$, $u_{\text{far}} = 0$ on $B_{r_0}(x_*)$, and u_{far} has the same uniform exterior regularity bounds as u on compact spacetime subsets away from (x_*, T).

Proof. Set $A = B_{2r_0}(x_*) \setminus B_{r_0}(x_*)$. Since $\nabla \cdot u = 0$,

$$\int_A \nabla \chi \cdot u \, dx = \int_{\mathbb{R}^3} \nabla \chi \cdot u \, dx = -\int_{\mathbb{R}^3} \chi \nabla \cdot u \, dx = 0.$$

For each time under consideration, the Bogovskii operator on the smooth annulus A gives $w(t) \in C_c^\infty(A)$ with the stated divergence. The exterior regularity hypothesis applies on the support of $\nabla \chi$. The two fields are divergence-free by construction. The identities on $B_{r_0}(x_*)$ follow from $\chi \equiv 1$ there and from the support condition on w. The asserted regularity is inherited from the exterior regularity hypothesis and the boundedness of the Bogovskii operator on the fixed annulus. □

Lemma 3.5 (Exterior fields vanish after blowup rescaling)**.** *Assume Assumption 3.3. Let $x_n \to x_*$ and $\lambda_n \downarrow 0$. Let $(u_{\text{ext}}, p_{\text{ext}})$ be a smooth exterior contribution supported in $\{|x - x_*| \geq r_0\}$, with $r_0 > 0$. Then*

$$\lambda_n u_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 s) \to 0$$

in $C^k(B_R \times I)$ for every fixed R, compact time interval I, and integer k. The associated pressure contribution converges, after subtracting constants depending on n and s, to zero in $L_s^\infty C_y^1(B_R)$.

Proof. For fixed R, the sets $x_n + \lambda_n B_R$ lie in $B_{r_0/2}(x_*)$ for all sufficiently large n, whereas u_{ext} is supported outside $B_{r_0}(x_*)$. This gives the velocity conclusion. The pressure p_{ext} is harmonic and uniformly smooth on the physical neighborhood swept out by $x_n + \lambda_n B_R$. If

$$Q_n(y, s) = \lambda_n^2 p_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 s),$$

then subtracting $\lambda_n^2 p_{\text{ext}}(x_n, t_n + \lambda_n^2 s)$ and using Taylor's theorem gives a difference bounded by

$$C_R \lambda_n^3 \|\nabla p_{\text{ext}}\|_\infty + C_R \lambda_n^4 \|D^2 p_{\text{ext}}\|_\infty,$$

which tends to zero. □

Corollary 3.6 (Profiles away from the blowup point are exterior)**.** *Let a profile have centers z_n satisfying $\liminf_{n\to\infty} |z_n - x_*| > 0$. Then the profile converges to zero in every blowup rescaling centered at x_*, and its pressure contribution converges to zero modulo constants. It is therefore an exterior regular profile, not a profile centered at the blowup point.*

Proof. Apply Lemma 3.5 with r_0 smaller than the limiting physical separation. □

4. Critical Tightness and Radiation

A sequence $(f_n) \subset L^3(\mathbb{R}^3)$ is critically tight if for every $\varepsilon > 0$ there is $R < \infty$ such that

$$\sup_n \int_{|x|>R} |f_n(x)|^3 dx < \varepsilon. \tag{7}$$

Lemma 4.1 (Compact sets in L^3 are tight)**.** *If $K \subset L^3(\mathbb{R}^3)$ is compact, then for every $\varepsilon > 0$ there is $R < \infty$ such that*

$$\sup_{f \in K} \int_{|x|>R} |f(x)|^3 dx < \varepsilon.$$

Proof. Choose a finite $\varepsilon^{1/3}/2$-net $f_1, \dots, f_N$ for K in L^3. Choose R so that each f_i has L^3-tail smaller than $\varepsilon^{1/3}/2$ outside B_R. The triangle inequality gives the claim. □

Proposition 4.2 (Compact part plus vanishing remainder implies tightness)**.** *Suppose*

$$f_n = h_n + r_n, \qquad h_n \in K \subset L^3(\mathbb{R}^3), \qquad \|r_n\|_3 \to 0,$$

where K is compact. Then (f_n) is critically tight.

Proof. The compact family (h_n) is tight by Lemma 4.1. Since $\|r_n\|_3 \to 0$, the tail of r_n is uniformly small for all sufficiently large n. Enlarging R to handle the finitely many remaining indices gives (7). □

Theorem 4.3 (Radiative tail criterion)**.** *Assume Assumption 2.2. Suppose that, after excluding perturbative profiles and exterior regular profiles, the critical data sequence has a compact L^3 part plus a remainder converging to zero in L^3. Then the sequence is critically tight. Therefore loss of critical tightness can occur only if a non-perturbative profile escapes to infinity, an exterior profile has not been excluded, the critical remainder does not vanish, or the profile expansion is incomplete.*

Proof. Apply Proposition 4.2 to the compact part and the vanishing remainder. The listed alternatives are the contrapositive, interpreted through the completeness clause in Assumption 2.2 and the exclusions in Corollary 3.2 and Lemma 3.5. □

5. Interaction Estimates on Compact Cylinders

The next assumption and lemmas isolate the nonlinear stability estimates needed to pass from a profile decomposition at one time to a decomposition of Navier–Stokes solutions on compact parabolic cylinders.

Assumption 5.1 (Local bounds and residual convergence)**.** Let $I \subset \mathbb{R}$ be compact and $R < \infty$. In a chosen rescaling, write the rescaled solution as

$$v_n = w_n + z_n$$

on $B_{2R} \times I$, where w_n is the nonlinear evolution of one chosen non-perturbative profile and z_n is the sum of all other profiles and the remainder in that rescaling. Assume

$$\sup_n \Big(\|w_n\|_{L^\infty_s L^3_y(B_{2R})} + \|w_n\|_{L^2_s H^1_y(B_{2R})} + \|z_n\|_{L^2_s H^1_y(B_{2R})} \Big) < \infty,$$

$$z_n \to 0 \quad \text{in } L^\infty_s L^3_y(B_{2R}),$$

and

$$\partial_s z_n - \nu \Delta z_n \to 0 \quad \text{in } L^1_s H^{-1}_y(B_R),$$

modulo the forcing terms associated with the remaining profiles. These forcing terms converge to zero in the same space. The pressure associated with interactions containing at least one factor z_n is the Calderon–Zygmund pressure determined by the corresponding interaction stress.

Lemma 5.2 (Cross-stress convergence)**.** *Under Assumption 5.1,*

$$w_n \otimes z_n + z_n \otimes w_n + z_n \otimes z_n \to 0 \quad \text{in } L^1_s L^2_y(B_R).$$

Proof. By Sobolev on B_{2R}, the local H^1 bounds imply boundedness of w_n and z_n in $L^2_s L^6_y(B_{2R})$. Hence

$$\|w_n \otimes z_n\|_{L^1_s L^2_y(B_R)} \le \|z_n\|_{L^\infty_s L^3_y(B_R)} \|w_n\|_{L^1_s L^6_y(B_R)} \to 0,$$

and the other two terms are identical. □

Lemma 5.3 (Pressure tails)**.** *Let $F_n \to 0$ in $L^1_s L^2_y(B_A)$ for every fixed A, and assume*

$$\sup_n \|F_n\|_{L^\infty_s L^{3/2}_y(\mathbb{R}^3)} < \infty.$$

If P_n is a pressure associated with F_n, namely

$$-\Delta P_n = \partial_i \partial_j F_{n,ij}$$

in $\mathbb{R}^3$, then for every $R < \infty$ there are functions $c_{n,R}(s)$ such that

$$P_n - c_{n,R} \to 0 \quad \text{in } L^1_s L^2_y(B_R), \qquad \nabla P_n \to 0 \quad \text{in } L^1_s H^{-1}_y(B_R).$$

Proof. Fix $A > 2R$. The local part is controlled by the L^2 boundedness of the Riesz transforms. For the far part, let Γ_{ij} be the pressure kernel and subtract

$$c_{n,R}(s) = \int_{|z|>A} \Gamma_{ij}(-z) F_{n,ij}(z,s)\, dz.$$

For $y \in B_R$ and $|z| > A$,

$$|\Gamma_{ij}(y-z) - \Gamma_{ij}(-z)| \le C_R |z|^{-4}.$$

Holder's inequality gives

$$\int_{|z|>A} |z|^{-4} |F_n(z,s)|\, dz \le C A^{-3} \|F_n(s)\|_{L^{3/2}}.$$

Taking $n \to \infty$ for the local part and then $A \to \infty$ gives the pressure convergence. The gradient convergence follows by testing against $H^1_0(B_R)$. □

Proposition 5.4 (Stability under vanishing secondary profiles)**.** *Under Assumption 5.1, replacing $v_n = w_n + z_n$ by w_n changes the Navier–Stokes equation on $B_R \times I$ by an error converging to zero in $L^1_s H^{-1}_y(B_R)$, after subtracting pressure constants depending on n and s.*

Proof. The linear residual of z_n tends to zero by assumption. The nonlinear error is

$$\nabla \cdot (w_n \otimes z_n + z_n \otimes w_n + z_n \otimes z_n),$$

which tends to zero in $L^1_s H^{-1}_y(B_R)$ by Lemma 5.2. The pressure error tends to zero by Lemma 5.3 applied to the interaction stress. □

6. Multibubbles

We now record the standard reductions of profiles with one limiting center and profiles with distinct limiting centers. They are consequences of critical orthogonality, local compactness of L^3, and the perturbative interaction estimates above.

Lemma 6.1 (Comparable profiles with one limiting center)**.** *Suppose finitely many profiles have the same limiting point and mutually comparable scales:*

$$\frac{\lambda_{j,n}}{\lambda_{1,n}} \to \rho_j \in (0,\infty), \qquad \frac{x_{j,n}-x_{1,n}}{\lambda_{1,n}} \to a_j.$$

In the scale $\lambda_{1,n}$ and center $x_{1,n}$, their sum converges locally, and globally when the profile convergence is global, to

$$\sum_j \rho_j^{-1}\phi^j\left(\frac{y-a_j}{\rho_j}\right).$$

Thus comparable profiles with one limiting center are a single profile in a common rescaling. If this sum has L^3 norm below ε_0, it is perturbative by Corollary 3.2.

Proof. The maps $f(y) \mapsto \rho^{-1}f((y-a)/\rho)$ are strongly continuous on $L^3(\mathbb{R}^3)$ for $\rho \in (0,\infty)$, $a \in \mathbb{R}^3$. This is immediate for smooth compactly supported functions and follows for general L^3 functions by density and Lemma 2.1. □

Lemma 6.2 (Larger-scale profiles vanish at the smaller scale)**.** *Let*

$$W_n(y) = \mu_n\phi(\xi_n + \mu_n y), \qquad \mu_n \to 0, \qquad \phi \in L^3(\mathbb{R}^3).$$

Then $W_n \to 0$ in $L^3(B_R)$ for every $R < \infty$.

Proof. By the change of variables $z = \xi_n + \mu_n y$,

$$\|W_n\|^3_{L^3(B_R)} = \int_{B(\xi_n,\mu_n R)} |\phi(z)|^3 dz.$$

The radii tend to zero, so the absolute continuity of the integral of $|\phi|^3$ gives the result. □

Lemma 6.3 (Regular larger-scale profiles give constant velocities)**.** *Let*

$$W_n(y) = \mu_n\phi(\xi_n + \mu_n y), \qquad \mu_n \to 0, \qquad \xi_n \to \xi,$$

and assume that ϕ is C^2 near ξ. Then for every $R < \infty$,

$$W_n(y) = \mu_n\phi(\xi_n) + \mu_n^2 D\phi(\xi_n)y + O_{L^\infty(B_R)}(\mu_n^3R^2).$$

The first term is a spatially constant velocity and can be absorbed by a Galilean change of frame; the remaining terms converge to zero locally.

Proof. Taylor's theorem at ξ_n gives the expansion. The Navier–Stokes equations are invariant under Galilean transformations, so a constant velocity may be absorbed into the moving spatial frame. □

Lemma 6.4 (Distinct centers decouple under blowup rescaling)**.** *Let $x_{p,n} \to x_p$, $x_{q,n} \to x_q$, $x_p \neq x_q$, and $\lambda_{p,n}, \lambda_{q,n} \downarrow 0$. In the rescaling centered at $x_{p,n}$ with scale $\lambda_{p,n}$, the q-profile has the form*

$$W_n(y) = \lambda_{p,n}(\lambda_{q,n})^{-1}\phi\left(\frac{x_{p,n}+\lambda_{p,n}y-x_{q,n}}{\lambda_{q,n}}\right).$$

Then $W_n \to 0$ in $L^3(B_R)$ for every $R < \infty$.

Proof. Changing variables gives

$$\|W_n\|^3_{L^3(B_R)} = \int_{B(c_n,\rho_n)} |\phi(z)|^3 dz,$$

where

$$c_n = \frac{x_{p,n}-x_{q,n}}{\lambda_{q,n}}, \qquad \rho_n = \frac{\lambda_{p,n}R}{\lambda_{q,n}}.$$

Since $|x_p - x_q| > 0$, $|c_n| \to \infty$, and $\rho_n/|c_n| \to 0$. The balls $B(c_n, \rho_n)$ escape to infinity; the L^1-tail of $|\phi|^3$ tends to zero. ☐

Lemma 6.5 (Distinct-center pressure profiles)**.** *In the setting of Lemma 6.4, suppose the associated profile pressure* Π *belongs to* $L^{3/2}(\mathbb{R}^3)$*. Its rescaled pressure contribution satisfies*

$$(\lambda_{p,n})^2(\lambda_{q,n})^{-2}\Pi\left(\frac{x_{p,n} + \lambda_{p,n}y - x_{q,n}}{\lambda_{q,n}}\right) \to 0$$

in $L^{3/2}(B_R)$ *for every* $R < \infty$.

Proof. The same change of variables as in Lemma 6.4 gives the integral of $|\Pi|^{3/2}$ over escaping balls. Since $\Pi \in L^{3/2}$, that integral tends to zero. ☐

Assumption 6.6 (Single-profile rigidity)**.** There is no single-profile blowup sequence arising after the preceding reductions and satisfying the corresponding compactness, local energy, pressure, and self-similar or stationary limiting hypotheses. In the backward self-similar case this is the theorem of Nečas–Růžička–Šverák [NRŠ96].

Theorem 6.7 (Single-center multibubble reduction)**.** *Assume Assumptions 2.2, 5.1 and 6.6. A finite multibubble configuration with one limiting center and a positive* L^3 *lower bound for each non-perturbative profile is incompatible with these assumptions. More precisely, after grouping comparable profiles, choosing the smallest scale, absorbing constant velocities by a Galilean change of frame, and using Proposition 5.4, the configuration reduces to a single-profile blowup sequence. This contradicts Assumption 6.6.*

Proof. By Lemma 2.5, only finitely many non-perturbative profiles are present. Comparable profiles at the same scale are a single profile by Lemma 6.1. At the smallest remaining scale, all larger-scale profiles with the same limiting center either vanish locally by Lemma 6.2 or, in the regular finite-offset case, are a constant velocity plus a locally vanishing error by Lemma 6.3. Constant velocities are absorbed by a Galilean change of frame. The nonlinear interaction and pressure errors are perturbative by Proposition 5.4. The smallest scale therefore yields a single-profile blowup sequence, which is excluded by Assumption 6.6. ☐

Theorem 6.8 (Distinct-center multibubble reduction)**.** *Assume Assumptions 2.2, 5.1 and 6.6. A finite multibubble configuration with at least two distinct limiting centers is incompatible with these assumptions, provided the associated pressure profiles and pressure interaction terms satisfy the local convergence stated in Lemma 6.5 and Lemma 5.3.*

Proof. Choose one limiting center and rescale at one of its non-perturbative profiles. Profiles centered at every other limiting point vanish locally by Lemma 6.4, and their associated pressure contributions vanish by Lemma 6.5. Mixed pressure and nonlinear interaction terms are perturbative by Proposition 5.4. The chosen limiting center therefore yields a single-profile blowup sequence, again contradicting Assumption 6.6. Since the chosen center was arbitrary, no such multibubble configuration exists. ☐

7. Main Theorem

Theorem 7.1 (Radiation and multibubble alternatives)**.** *Let* (u, p) *be a suitable weak solution near* (x_*, T)*, and let* (f_n) *be a bounded critical sequence obtained from the rescalings* (1)*. Assume:*

(i) *the critical profile decomposition and completeness statement in Assumption 2.2;*
(ii) *Kato small-data perturbation theory, as in Theorem 3.1;*
(iii) *exterior regularity away from* (x_*, T)*, as in Assumption 3.3;*
(iv) *nonlinear profile stability on compact parabolic cylinders, as in Assumption 5.1;*

(v) *the single-profile Liouville or rigidity hypothesis in Assumption 6.6.*

Then every non-perturbative critical concentration profile belongs to a single limiting point in $\mathbb{R}^3$ *and a single asymptotic scale, after comparable profiles are grouped. In particular, a radiative loss of critical tightness, a single-center scale cascade, or a distinct-center multibubble can occur only if one of the five assumptions above fails.*

Proof. Profiles with L^3 norm at most ε_0 are perturbative by Corollary 3.2. Profiles separated from x_* are exterior regular by Lemma 3.5. If, after these exclusions, the remaining contribution is compact in L^3 up to a vanishing remainder, then there is no radiative loss by Theorem 4.3; hence any radiative loss is a failure of profile completeness, remainder smallness, or exterior exclusion.

The remaining non-perturbative profiles are finite in number by Lemma 2.5. Comparable profiles at the same point and scale combine into one profile by Lemma 6.1. Strict single-center scale separation is excluded by Theorem 6.7. Distinct limiting centers are excluded by Theorem 6.8. Thus no radiative or multibubble alternative remains unless one of the stated assumptions fails. □

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